

XIAOYU (OLIVIA) YANG

oliviayang547@gmail.com | (608) 471-2259 | [LinkedIn](#)

EDUCATION

University of Chicago

Master of Science, Applied Data Science

University of Wisconsin-Madison

Bachelor of Science

Majors: Statistics & Data Science; Minors: Math & Computer Science

Chicago, IL

Sep 2025 - Expected Dec 2026

Madison, WI

Sep 2021 - May 2025

SKILLS

Language: Python (Scikit-Learn, NumPy, Pandas, Matplotlib, PyTorch), SQL, R

Visualization: Tableau, Power BI

Tech Stack: AWS, Microsoft Azure, Google Cloud Platform, LangChain, LiteLLM, Machine Learning (XGBoost, CNN, MLP), Prompt Engineering, Agent Coding, Hadoop

Relevant Courses: Next-Generation Natural Language Processing, Big Data and Cloud Computing, Machine Learning

WORK EXPERIENCE

CAS (Chemical Abstracts Services)

Columbus, OH

Data Scientist Intern

June 2026 - Expected Sep 2026

- Built to an internal **LLM** evaluation pipeline (ChemBench) benchmarking 20+ models (**Claude**, **GPT**, etc.) across chemistry, image, **NER**, and textbook question domains, covering model integration and evaluation methodology
- Integrated OpenAI models into the evaluation pipeline across three reasoning-effort levels, migrating the LLM integration layer from **LangChain** to **LiteLLM** to support **Azure's** Responses API alongside Bedrock
- Eliminating empty responses across all effort levels by identifying a validity drop (73% to 33%) in high-reasoning-effort model outputs and **tuning max output tokens**
- Improving leaderboard reliability by 16.7% by implementing a 3-judge-averaging approach after identifying and flagging two sources of judge-score variance (run-to-run LLM-as-judge noise, cross-provider judge bias)

AI Industry Impact Analysis, University of Chicago

Chicago, IL

Class Project, Next-Generation Natural Language Processing

Mar 2026

- Identified industries and companies most impacted by AI by analyzing ~200,000 AI-related news articles (2022–2026) using **NLP**, and uncovered that Legal was the only consistently negative industry while Computer & Mathematical and Transportation trended positive
- Built a data cleaning pipeline that removed crawl artifacts, duplicate articles via **content fingerprinting**, and non-news AI mentions, then applied **keyword-based relevance filtering** across AI, industry, workforce, and impact terms to narrow the raw corpus down to 110,000 relevant articles
- Applied **BERTopic** (UMAP + HDBSCAN + c-TF-IDF) to discover 188 topics across 110K articles, then used **LLaMA 3.1-8B** to map topics to Goldman Sachs occupation categories, surfacing 11 industries with AI-related discourse
- Performed **named entity recognition** (spaCy) and **sentiment analysis** to track industry- and company-level sentiment toward AI, and synthesized findings into concrete success and failure factors for AI adoption, such as framing AI as augmentation rather than replacement and the role of infrastructure readiness versus regulatory and copyright uncertainty

Berry Genomics

Beijing, China

Artificial Intelligence Algorithm Intern

Jul 2024 - Sep 2024

- Increased data readiness for **Genome-Wide Association Studies (GWAS)** by 30% and enabled accurate detection of cardiac-related genetic variants by building a genomic **data cleaning pipeline** for 1,000+ VCF samples.
- Conducted GWAS on prenatal datasets of 1,000+ samples, successfully identifying key genetic markers associated with cardiac diseases despite limited sample variation.
- Implemented **LLMs (Qwen, LLaMA)** to analyze gene-phenotype correlations by integrating two datasets, enhancing the model's predictive accuracy for identifying cardiac disease-related variants by 22%.

RESEARCH EXPERIENCE

Lu Lab, UW-Madison

Madison, WI

Research Assistant

Jan 2023 - May 2025

- Assist supervisors with data processing, analysis, and model construction to contribute to genetics journal publications of [PENGUIN](#) (adjusting for genetic confounding) and [PUMAS](#) (ensemble learning for PRS) softwares, [M-Learner model](#) (causal transfer learning in clinical trials).